



KNOWLEDGE BASE

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NSX 9 OVA Deploy PowerCLI Import-VApp vs ovftool

Verified-reliable deployment path when ovftool 5.0 silently drops vAppConfig on vSAN-target deploys. Documented end-to-end from a 2026-05-06 incident.

NSX 9

POWERCLI

OVFTOOL

VAPPCONFIG

IMPORT-VAPP

VSAN

VCF 9.0

VMware Cloud Foundation

NSX 9 OVA Deploy — PowerCLI Import-VApp vs ovftool

Prepared by: Virtual Control LLC **Date:** May 6, 2026 **Document Version:** 1.0 **Classification:** Internal — Knowledge Base **Scope:** VCF 9.0.1 nested lab, NSX Manager redeploy

1. Executive Summary

On 2026-05-06 during a planned NSX Manager rebuild, **ovftool 5.0.0 build-24927197 deployed the NSX 9 OVA twice with Completed successfully exit code 0, but produced vAppConfig=null on the resulting VM both times.** OVA properties (`nsx_ip_0` , `nsx_hostname` , etc.) were silently dropped, and the appliance fell back to OVA defaults: DHCP-assigned IP `192.168.1.161` and hostname `nsx-unified-appliance` — not the configured `192.168.1.71` and `nsx-manager.lab.local` .

The same `ovftool` flags worked on a 2026-04-25 redeploy attempt with the same version and same OVA file. The 2026-05-06 regression was not reproducible to a single root cause but the failure was definitive: vCenter API confirmed `vAppConfig: null` and the VMX file contained no `guestinfo.ovfEnv` keys after either `ovftool` run.

The third attempt switched to **PowerCLI Import-VApp with OvfConfiguration** — this uses vCenter's `Folder.ImportVApp` API path which provably preserves vApp configuration. The deploy succeeded on the first attempt: `vAppConfig: NOT NULL, properties=27` , including `nsx_ip_0=192.168.1.71` . After bumping resources via `Set-VM` (which preserves vAppConfig), the appliance booted with the correct static IP and hostname, reaching cluster `mgmt: STABLE / control: STABLE` .

This document captures the verified-reliable PowerCLI pattern, the diagnostic approach that confirmed `ovftool` was at fault, and the operational guardrails that prevent the silent-failure mode from recurring.

2. Symptom — What ovftool Did Wrong

2.1 The misleading success

```
ovftool 5.0.0 build-24927197
$OVFTOOL --acceptAllEulas --noSSLVerify --skipManifestCheck \
  --allowExtraConfig --allowAllExtraConfig \
  --diskMode=thin --deploymentOption=small \
  --datastore=vcenter-cl01-ds-vsan01 \
  '--net:Network 1=pg-vm-mgmt' \
  --name=nsx-manager --powerOffTarget \
  '--prop:nsx_role=NSX Manager' \
  '--prop:nsx_passwd_0=<nsx-admin-password>' \
  '--prop:nsx_cli_passwd_0=<nsx-cli-password>' \
  '--prop:nsx_cli_audit_passwd_0=<nsx-cli-audit-password>' \
  --prop:nsx_hostname=nsx-manager.lab.local \
  --prop:nsx_gateway_0=192.168.1.1 \
  --prop:nsx_ip_0=192.168.1.71 \
  --prop:nsx_netmask_0=255.255.255.0 \
  --prop:nsx_dns1_0=192.168.1.230 \
  --prop:nsx_domain_0=lab.local \
  --prop:nsx_ntp_0=192.168.1.230 \
```

```
--prop:nsx_isSSHEnabled=True \  
--prop:nsx_allowSSHRootLogin=True \  
'/i/VCF-Depot/PROD/COMP/NSX_T_MANAGER/nsx-unified-appliance-9.0.1.0.24952114.ova' \  
'vi://administrator%40vsphere.local:<vcenter-administrator-password-url-encoded>@192.168.1.69/vcenter-  
dc01/host/vcenter-cl01/esxi04.lab.local'
```

Output:

```
Disk progress: 1%...100% Transfer Completed  
Warning: The manifest is present but user flag causing to skip it  
Completed successfully
```

Exit code **0**. No errors logged. VM appears in vCenter inventory.

2.2 The actual deployed state

```
$ plink ... root@esxi04 'vim-cmd vmsvc/get.config <vmid> | grep -E "vAppConfig|numCPU|memoryMB"'  
numCPU = 4, # ← OVA "small" defaults, NOT customized  
memoryMB = 16384, # ← OVA defaults, NOT customized  
vAppConfig = (vim.vApp.VmConfigInfo) null, # ← THIS IS THE BUG
```

```
$ plink ... root@esxi04 'cat /vmfs/.../nsx-manager.vmx | grep -i "guestinfo\|nsx_"'  
(empty output – no guestinfo.ovfEnv, no nsx_* keys)
```

The deployed VM had `vAppConfig: null` and no OVF-environment `guestinfo`. NSX appliance firstboot script could not read the configured properties, so it fell to defaults: DHCP for IP, factory hostname `nsx-unified-appliance`.

2.3 What the user saw

After power-on, VMware Tools reported the guest IP as `192.168.1.161` (DHCP assignment), not the configured `192.168.1.71`. The vCenter UI also showed the VM hostname as `nsx-unified-appliance`. Web access to `192.168.1.71` returned no response.

3. Why It Happened (Honest Assessment)

The exact root cause was **not isolated** during the incident. Possibilities considered:

Hypothesis	Evidence
Mid-deploy <code>vim-cmd vmsvc/reload</code> raced with ovftool's vCenter API call	First failure occurred after a reload was issued mid-flight; second failure occurred WITHOUT any reload, so this was ruled out
Single-VM OVA (not vApp) goes through <code>Folder.ImportVm</code> path which doesn't preserve vApp config	Plausible per ovftool/vCenter API behavior, but the same path worked on 2026-04-25
Cached state in vCenter from an orphaned VM record	Orphan was deleted before each retry, so this was ruled out
ovftool 5.0 regression specific to vSAN target	Not confirmed; same version was used on 2026-04-25 successfully

Honest position: the failure is real and reproducible (twice in the same session), but the trigger is not understood. The mitigation (PowerCLI Import-VApp) is verified to work regardless. The diagnostic approach in §5 catches the failure in seconds rather than waiting for boot.

4. The Verified-Reliable Path: PowerCLI Import-VApp

4.1 Why Import-VApp works

PowerCLI's `Import-VApp` cmdlet calls vCenter's `Folder.ImportVApp` API path explicitly. This is the same API used by the vSphere Web Client's "Deploy OVF" wizard, and it is documented to preserve vApp configuration regardless of whether the OVA contains a single `VirtualSystem` or a `VirtualSystemCollection`. `ovftool`'s behavior depends on which API path it chooses internally based on the OVA structure and target type, and that decision can be opaque in error cases.

4.2 Working PowerCLI script (verified end-to-end 2026-05-06)

```
Set-Location C:\
Import-Module VMware.PowerCLI -ErrorAction SilentlyContinue
Set-PowerCLIConfiguration -InvalidCertificateAction Ignore -Confirm:$false -Scope Session | Out-Null
$cred = New-Object System.Management.Automation.PSCredential(
    'administrator@vsphere.local',
    (ConvertTo-SecureString 'vc-password' -AsPlainText -Force))
Connect-VIServer -Server 192.168.1.69 -Credential $cred -ErrorAction Stop | Out-Null

$ova = 'I:\VCF-Depot\PROD\COMP\NSX_T_MANAGER\nsx-unified-appliance-9.0.1.0.24952114.ova'
$vmhost = Get-VMHost -Name esxi04.lab.local
$ds = Get-Datastore -Name vcenter-cl01-ds-vsan01
$net = Get-VDPortgroup -Name pg-vm-mgmt

# Read OVF config from OVA – populates the Common section with all OVA properties
$ovfConfig = Get-OvfConfiguration -Ovf $ova

# Set OVA properties (these would have been --prop:* in ovftool)
$ovfConfig.Common.nsx_role.Value = 'NSX Manager'
$ovfConfig.Common.nsx_passwd_0.Value = '<root-pw>'
$ovfConfig.Common.nsx_cli_passwd_0.Value = '<admin-pw>'
$ovfConfig.Common.nsx_cli_audit_passwd_0.Value = '<audit-pw>'
$ovfConfig.Common.nsx_hostname.Value = 'nsx-manager.lab.local'
$ovfConfig.Common.nsx_gateway_0.Value = '192.168.1.1'
$ovfConfig.Common.nsx_ip_0.Value = '192.168.1.71'
$ovfConfig.Common.nsx_netmask_0.Value = '255.255.255.0'
$ovfConfig.Common.nsx_dns1_0.Value = '192.168.1.230'
$ovfConfig.Common.nsx_domain_0.Value = 'lab.local'
$ovfConfig.Common.nsx_ntp_0.Value = '192.168.1.230'
$ovfConfig.Common.nsx_isSSHEnabled.Value = $true
$ovfConfig.Common.nsx_allowSSHRootLogin.Value = $true

# Network mapping (the OVA defines a "Network 1" entry; mapping property is "Network_1")
$ovfConfig.NetworkMapping.Network_1.Value = $net

# Deployment option (OVA has "small" defined; matches --deploymentOption=small)
$ovfConfig.DeploymentOption.Value = 'small'

# Deploy – this calls Folder.ImportVApp internally
$vm = Import-VApp -Source $ova `
    -OvfConfiguration $ovfConfig `
    -Name 'nsx-manager' `
    -VMHost $vmhost `
    -Datastore $ds `
    -DiskStorageFormat Thin `
    -Force

# CRITICAL: verify vAppConfig was actually populated BEFORE proceeding
$cfg = $vm.ExtensionData.Config.VAppConfig
if ($cfg -eq $null) {
    throw "vAppConfig is NULL after Import-VApp – abort, do not power on"
}
"vAppConfig populated, $($cfg.Property.Count) properties, nsx_ip_0=$((($cfg.Property | Where-Object Id -eq 'nsx_ip_0').Value))"
```

```
# Resources: bump from "small" defaults (4/16) to lab-required (6/48)
# Set-VM uses vCenter reconfig API which PRESERVES vAppConfig
Set-VM -VM $vm -NumCpu 6 -MemoryGB 48 -Confirm:$false | Out-Null

# Re-verify vAppConfig STILL non-null after Set-VM
$cfg = $vm.ExtensionData.Config.VAppConfig
if ($cfg -eq $null) { throw "vAppConfig dropped during Set-VM - abort" }

Start-VM -VM $vm | Out-Null
Disconnect-VIServer -Confirm:$false | Out-Null
```

4.3 The two `vAppConfig` checks are critical

Check	Why
Immediately after <code>Import-VAp</code> <code>p</code> returns	If null here, the deploy itself failed silently — do not power on
Immediately after <code>Set-VM -Nu</code> <code>mCpu/-MemoryGB</code>	Verifies the resource bump preserved vApp config (it should — <code>Set-VM</code> uses the same reconfig API path; documented as preserving vApp config)

If either check fails, the appliance will fall back to defaults at power-on. The check is cheap (~milliseconds) and catches the silent-failure mode before any boot time is wasted.

5. Diagnostic — Detect ovftool's Silent Failure in Seconds

If you must use ovftool (e.g., not in a PowerCLI environment), validate immediately after deploy:

```
# From the host running the new VM
$ plink ... root@<host> 'vim-cmd vmvc/get.config <vmid> | grep -E "vAppConfig|nsx_ip|nsx_hostname"'
```

Expected (success): `vAppConfig` is a populated `vim.vApp.VmConfigInfo` object with `Property` array.

Failure mode: `vAppConfig = (vim.vApp.VmConfigInfo) null`, — this is the silent-failure signature. Power-on will use OVA defaults.

Or via PowerCLI:

```
$vm = Get-VM -Name nsx-manager
$vm.ExtensionData.Config.VAppConfig # null = bad
$vm.ExtensionData.Config.VAppConfig.Property | Format-Table Id, Value
```

6. Step-by-Step from the 2026-05-06 Incident

This is the actual sequence that resolved the issue, with timestamps and outputs.

6.1 First ovftool attempt (failed silently)

Time UTC	Event
12:38	ovftool ... vi://...esxi04.lab.local — "Duplicate name 'nsx-manager' on target" (orphan from prior VM destroy)
12:43	PowerCLI: Get-VM nsx-manager | Where ConnectionState='orphaned' | Remove-VM — orphan removed
12:56	ovftool retry — deploy started
12:57	Operator edited nsx-manager.vmx to bump numvcpus="6" and memSize="49152", ran vim-cmd vmsvc/reload . This is the moment that may have raced with the deploy.
13:43	ovftool exit 0, "Completed successfully"
13:43	vim-cmd vmsvc/get.config 4 | grep vAppConfig → null . SILENT FAILURE.

6.2 Second ovftool attempt (failed identically, no race this time)

Time UTC	Event
13:50	First VM destroyed via vim-cmd vmsvc/destroy 4
13:51	Orphan record removed via PowerCLI
13:52	ovftool retry — NO mid-deploy edits this time
14:38	ovftool exit 0, "Completed successfully"
14:39	vim-cmd vmsvc/get.config <new-vmid> | grep vAppConfig → null . SAME FAILURE.

The second failure with no operator interference confirmed the issue was not the mid-deploy reload.

6.3 OVA descriptor inspection (ruled out property name issues)

```
$OVFTOOL --hideEula --schemaValidate=false --allowExtraConfig --allowAllExtraConfig "$OVA"
# Returned all 24 user-configurable properties matching exactly what we passed via --prop:*
# Confirmed property names were correct
```

6.4 PowerCLI Import-VApp (worked first try)

Time UTC	Event
14:42	Second failed VM destroyed
14:42	Orphan removed
14:43	PowerCLI Import-VApp script launched
~15:35	Import completed
15:35	\$vm.ExtensionData.Config.VAppConfig → NOT NULL, 27 properties , nsx_ip_0 = 192.168.1.71 confirmed
15:35	Set-VM -NumCpu 6 -MemoryGB 48 → vAppConfig still NOT NULL after resize
15:35	Start-VM — powered on
~15:55	API responding on https://192.168.1.71/api/v1/cluster/status → mgmt: STABLE, control: STABLE

Total elapsed for the working PowerCLI path: ~50 minutes from launch to STABLE cluster (most of it ovftool-equivalent OVA upload time over LAN).

7. Lessons Learned

#	Lesson
L-1	ovftool 5.0.0 build-24927197 can silently drop vAppConfig on VCF 9 OVAs targeting vSAN. Exit code 0 is not sufficient validation.
L-2	Always verify vAppConfig is non-null IMMEDIATELY after deploy, before powering on. The check is one PowerShell line or one vim-cmd grep.
L-3	PowerCLI Import-VApp + Get-OvfConfiguration + Set-VM is the verified-reliable triplet for VCF 9 OVA deploys on vSAN. Use it over ovftool when this failure mode has been observed.
L-4	Never edit a VM's VMX or run vim-cmd vmsvc/reload while ovftool is uploading. Even if the failure is reproducible without it, the race window adds risk for no benefit.
L-5	Resource customization (vCPU/RAM bumps) belongs AFTER deploy completion via Set-VM (vCenter reconfig API), not via direct VMX edit. Set-VM preserves vAppConfig; VMX edit + reload may not.
L-6	The OVA's "small" deployment option produces 4 vCPU / 16 GB defaults. For lab nested ESXi running NSX Manager, the verified-required is 6 vCPU / 48 GB (per feedback_nsx_manager_dedicated_host.md history of RCU stalls under 16 GB).

8. Cross-References

- NSX-Manager-Cold-Start-Troubleshooting-Report.md Addendum D — the broader 2026-05-06 NSX rebuild context
- feedback_nsx_manager_dedicated_host.md — co-tenancy + resource sizing rule
- feedback_no_assumptions.md — the verify-before-claim rule that drove the empirical diagnosis
- VMware ovftool 5.0 documentation — for the upstream behavior
- VMware PowerCLI documentation — Import-VApp and Get-OvfConfiguration cmdlet references

9. Document Information

Field	Value
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Field	Value
OVA File	nsx-unified-appliance-9.0.1.0.24952114.ova
Failure Reproducibility	2 of 2 attempts on 2026-05-06; success rate of 1 of 1 with PowerCLI alternative

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